

Reproduction of microsoft/BitNet: BitNet b1.58 2B CPU inference throughput

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Independent reproduction · Artifact evaluation · Natural language processing

Abstract

Built bitnet.cpp at the pinned commit with setup_env.py (clang 18, i2_s kernels) and ran the README benchmark utils/e2e_benchmark.py on the official BitNet-b1.58-2B-4T GGUF: 512-token prompt processing and 128-token generation, batch 1, 8 threads, CPU only, 5 repetitions. At this commit e2e_benchmark.py exits with status 1 even on success, so success was judged by its result table. Intel Xeon W-2145.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	BitNet b1.58 2B CPU through...	2026-10-10	1	1	Signed by the K-Veritas server

2 Results

r1: BitNet b1.58 2B CPU throughput

Run	Metric	Final value	Step
1	pp512_tokens_per_second	23.65	
1	pp512_stddev	0.06	
1	tg128_tokens_per_second	24.7	
1	tg128_stddev	0.03	

Run	Inline claim	Value	Phase
1	pp512_tokens_per_second	23.65	
1	tg128_tokens_per_second	24.7	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	157.1 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	fe98743f8055023f804a73da8ca5feb34c0362996d48eab351c34b7cdfd3d8fa	3.3 MB
r1.kvbundle.zip	baac73e619227272be84dac4fe36b591485d452c157195d3ab63b6d16abeb3b6	186 KB

5 Links and licence

Paper: <https://arxiv.org/abs/2410.16144>

Reproduction of: <https://github.com/microsoft/BitNet/tree/0b341e582afbf9e1011f24744b554c96a3477eb5>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: 1-bit-llm, bitnet, cpu-inference, quantization.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (BitNet b1.58 2B CPU throughput), kv:2610.00031v1/r1. Data hash 926858f527e7224ce853e274f57f1916031cf2946e10b05df723ba82d0f5d13d.
<https://kveritas.org/records/2610.00031v1/r1.pdf>